Tender Ref ID: 996686 | Notice No: 21/MD/CE(ELE)AGICL/2026, Dt.28.09.2026
Estimated Contract Value: Rs. 3,37,90,33,865/- | Bid Closing: 30/10/2026 03:00 PM
File: SL-288-LA-_ELV_Tech_Spec.pdf | Portal: https://tender.apeprocurement.gov.in

SPECIFICATION CLAUSE 5.0 · INTEGRATED EXTRA LOW VOLTAGE (ELV) INFRASTRUCTUR

1. STRUCTURED CABLING SYSTEM (SCS):
? Scope: Complete category 6A F/UTP structured horizontal cabling and OM4 10-Gigabit optical fiber backbone.
? Standards: ANSI/TIA-568-C.2, ISO/IEC 11801 Class EA standards for 500 MHz channel transmission.
? All jack modules, patch panels, and patch cords shall be component-rated and sourced from a single OEM.
? 25-Year Application Assurance & Product Warranty from OEM directly in the name of AGICL.

2. IP VIDEO SURVEILLANCE & AI ANALYTICS:
? Coverage: High-definition IP cameras for all perimeter gates, public galleries, lobbies, assembly corridors, and entry/exit portals.
? Resolution: Minimum 4K UHD (3840x2160) for outdoor perimeter; 5MP WDR dome cameras with IR illumination for indoor spaces.
? Storage: RAID 6 enterprise SAN/NAS storage array supporting 90 days continuous recording at 25 fps.
? AI Analytics: Automated perimeter intrusion detection, left-object alert, license plate recognition (ANPR), and facial recognition at VVIP gates.

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SPECIFICATION CLAUSE 5.2 · ACCESS CONTROL & PUBLIC ADDRESS VOICE ALARM

3. INTEGRATED ACCESS CONTROL SYSTEM (ACS):
? Controllers: 32-bit distributed IP controllers with encrypted OSDP v2 communication to biometric/smartcard readers.
? Readers: Contactless Smart Card (Mifare DESFire EV3) + 3D Facial Recognition terminals at Chamber doors.
? Fail-Safe Integration: Complete hardwired interface with Fire Alarm Control Panel (FACP) to release all magnetic locks upon alarm.

4. PUBLIC ADDRESS & VOICE ALARM (PAVA):
? System shall comply with EN 54-16 (Control & Indicating Equipment) and EN 54-24 (Loudspeakers).
? Dual-redundant A/B loudspeaker circuits per acoustic zone to ensure intelligibility during partial circuit failure.
? STI (Speech Transmission Index) of not less than 0.55 across all public areas, press boxes, and legislative chambers.

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SPECIFICATION CLAUSE 5.3 · BUILDING MANAGEMENT SYSTEM (BMS) & IOT INTEGRATI

5. ENTERPRISE BUILDING MANAGEMENT SYSTEM (BMS):
? Architecture: Open-protocol native BACnet/IP building automation network integrating HVAC, Electrical, and Fire systems.
? Control Engine: High-performance DDC controllers monitoring chiller COP, AHU VFD modulation, and CO2 indoor air quality levels.
? Energy Management: Real-time energy sub-metering dashboard compliant with GRIHA 5-Star benchmarks.
? Cybersecurity: IEC 62443 cybersecurity standard compliance with encrypted TLS 1.3 web interface.